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ABSTRACT

The increasing problem of food wastage alongside rising food insecurity highlights the need for efficient and transparent food redistribution systems. A significant amount of surplus edible food from restaurants, hotels, and canteens is discarded daily due to the lack of proper coordination between donors and organizations that serve people in need. Existing centralized systems often suffer from inefficiency, lack of trust, and poor traceability, leading to delays and food spoilage.

The **FoodTrust** project aims to address these challenges by developing a **decentralized food donation and traceability platform** using blockchain, Internet of Things (IoT), and artificial intelligence (AI). The system enables donors to list surplus food, NGOs to accept and collect donations, and administrators to monitor operations through dedicated dashboards. IoT sensors are used to monitor food conditions such as temperature and humidity, while AI-based models predict food freshness to ensure safe consumption. Blockchain technology ensures transparency, immutability, and trust by securely recording donation, pickup, and delivery transactions, along with a token-based incentive mechanism to encourage participation.

The proposed system improves coordination between stakeholders, reduces food wastage, enhances food safety, and ensures accountable distribution of donated food. FoodTrust contributes toward building a sustainable and transparent food redistribution ecosystem and supports social welfare initiatives aimed at reducing hunger and promoting responsible resource utilization.

Keywords: Decentralized food donation, Blockchain-based traceability, IoT food monitoring, AI freshness prediction, Food waste management, Smart redistribution system, Transparency, Sustainability.

1. INTRODUCTION

Food wastage and food insecurity are major global challenges affecting both developed and developing countries. Large quantities of edible food are wasted every day by restaurants, hotels, event organizers, and food service institutions due to improper planning, lack of coordination, and limited redistribution mechanisms. At the same time, a significant portion of the population struggles with hunger and malnutrition, highlighting a serious imbalance in food availability and distribution. Addressing this issue is critical to achieving sustainable development and improving social welfare.

Despite the presence of food donation initiatives, many existing systems rely on centralized platforms that often lack transparency, real-time monitoring, and trust among stakeholders. Donors are frequently uncertain whether their food reaches the intended beneficiaries safely, while non-governmental organizations (NGOs) face difficulties in timely pickups, freshness verification, and efficient distribution. Additionally, the absence of reliable traceability and accountability increases the risk of food spoilage and misuse.

The **FoodTrust** project is proposed to overcome these limitations by introducing a **decentralized food donation and traceability system** that integrates **blockchain, Internet of Things (IoT), and artificial intelligence (AI)** technologies. The project aims to create a transparent and efficient ecosystem where surplus food can be donated, tracked, and distributed safely in real time. IoT sensors are used to monitor food storage conditions, while AI-based models predict food freshness to support safe consumption decisions. Blockchain technology ensures secure and immutable records of all transactions, enhancing trust among donors, NGOs, and administrators.

The primary objective of this project is to reduce food wastage while improving access to safe food for underprivileged communities. The scope of the work includes designing a user-friendly platform for multiple stakeholders, implementing real-time monitoring and predictive analysis, and ensuring transparent operations through decentralized mechanisms. FoodTrust aligns with societal needs by promoting sustainability, social responsibility, and technological innovation, and it contributes toward building a reliable and scalable solution for food redistribution systems in the future

2. LITERATURE REVIEW

A literature review provides an overview of existing research and technological solutions related to food waste management, food donation systems, and the use of emerging technologies such as blockchain, Internet of Things (IoT), and artificial intelligence (AI). It helps identify current approaches, methodologies, and limitations, thereby highlighting the research gap addressed by the proposed FoodTrust system.

Paper 1: Blockchain-Based Food Supply Chain Management

Publication Year: 2021

This study explores the application of blockchain technology in food supply chain management to improve transparency, traceability, and trust among stakeholders. The paper discusses how immutable ledgers can record food movement from source to destination, reducing fraud and improving accountability. While the system enhances traceability, it mainly focuses on large-scale supply chains and does not address real-time food donation or redistribution. The absence of IoT-based freshness monitoring limits its effectiveness for handling perishable surplus food.

Paper 2: Smart Food Waste Management Using IoT

Publication Year: 2020

The paper proposes an IoT-based food waste management system using sensors to monitor food quantity and condition in smart bins. It emphasizes real-time data collection for reducing waste in restaurants and households. Although the system effectively monitors food status, it relies on centralized data storage and lacks secure data sharing mechanisms. The study does not provide solutions for transparent donation tracking or trust-building among donors and NGOs.

Research Gap Identification

From the reviewed literature, it is evident that existing systems address food waste monitoring, blockchain-based traceability, or AI-driven analysis independently. However, there is a lack of **integrated solutions** that combine **blockchain for transparency**, **IoT for real-time food condition monitoring**, and **AI for freshness prediction** within a decentralized food donation ecosystem. The **FoodTrust** project aims to bridge this gap by providing a unified platform that ensures safe, transparent, and efficient redistribution of surplus food.

3. OBJECTIVE

The primary objective of the **FoodTrust** project is to develop a **decentralized, transparent, and efficient food donation and redistribution system** that minimizes food wastage and improves access to safe food for underprivileged communities. The project aims to leverage modern technologies to ensure accountability, trust, and real-time monitoring throughout the food donation lifecycle.

The specific objectives of the project are as follows:

- **Donor & NGO Coordination Platform** — Design an intuitive platform that enables donors to easily list surplus food and facilitates timely coordination between donors and NGOs for seamless food pickup and delivery.

- **IoT-Driven Food Safety Monitoring** — Integrate IoT sensors (temperature, humidity) to continuously monitor food storage conditions, combined with AI-based predictive models to assess food freshness and spoilage risk in real time.

- **Blockchain-Powered Transparency & Traceability** — Leverage blockchain technology to ensure end-to-end transparency and traceability of food donations, with smart contracts automating verification processes and incentive mechanisms for stakeholders.

- **Waste Reduction & Safe Redistribution** — Collectively reduce food wastage and uphold food safety standards throughout the redistribution process by unifying all platform components into a cohesive, user-friendly experience for donors, NGOs, and recipients alike.Overall, the FoodTrust project aims to create a scalable, reliable, and socially impactful solution that supports sustainable food management and contributes to hunger reduction initiatives.

4. Project Requirements

Project requirements define the essential resources and conditions needed to successfully design, develop, and deploy the FoodTrust system. These requirements are categorized into functional requirements, non-functional requirements, hardware requirements, and software requirements to ensure smooth operation and achievement of project objectives.

Functional Requirements

- The system should allow donors to register and submit details of surplus food available for donation.
- NGOs should be able to view available donations, accept pickup requests, and confirm delivery.
- The system must track food donation status from submission to final distribution.
- IoT sensors should collect environmental data such as temperature and humidity to monitor food conditions.
- AI models should analyze sensor data to predict food freshness and spoilage risk.
- Blockchain should securely record donation, pickup, and delivery transactions.
- Smart contracts should automate verification and incentive distribution.
- Administrators should be able to monitor system activity, generate reports, and manage users.

Non-Functional Requirements

- The system should provide high availability and reliable performance.
- Data security and privacy must be ensured through secure authentication and encrypted transactions.
- The platform should be scalable to support increasing numbers of users and donations.
- The user interface should be intuitive and easy to use for all stakeholders.
- The system should support real-time updates and notifications.

Following are the minimum hardware and software requirements for building this project:

Hardware:

Component	Specifiacation
Processor	Intel core i5- 8250U@1.60GHz or higher

RAM	8 GB or 16GB and more
Memory	8 GB and More

Software:

Category	**Specification**
Operating System	LINUX, WINDOW 10, WINDOWS 11
Programming Languages	TypeScript, JavaScript
Frameworks & Libraries	React, Node.js, Express.js
Database	Firebase
Web Browser	Google Chrome, Microsoft Edge, Safari, Mozilla Firefox
Text Editor	Visual Studio Code and Sublime Text editor
API Testing Tool	Postman
Documentation Tool	Microsoft Word

5. TECHNOLOGIES

This section describes the technologies and tools used in the development of the **FoodTrust** project. The selected technologies provide a strong technical foundation to ensure scalability, transparency, real-time monitoring, and efficient system performance.

Frontend

React: React is a JavaScript library used for building interactive and responsive user interfaces. It follows a declarative programming approach, allowing the application to update and render components efficiently based on changes in data. In FoodTrust, React is used to develop intuitive dashboards for donors, NGOs, and administrators, enabling smooth navigation and real-time data visualization.

Tailwind CSS: Tailwind CSS is a utility-first CSS framework that enables rapid UI development with consistent styling. It helps in creating clean, responsive, and user-friendly interfaces. In FoodTrust, Tailwind CSS is used to design visually appealing dashboards and ensure cross-device compatibility..

Backend

Express: Express.js is a lightweight and flexible web application framework for Node.js used to build RESTful APIs. It simplifies routing, middleware management, and API development. FoodTrust uses Express.js to manage donation workflows, user interactions, and data exchange between the frontend, database, and blockchain services.

Node.js: Node.js is an open-source, cross-platform JavaScript runtime environment that allows server-side execution of JavaScript code. It uses an event-driven, non-blocking I/O model, making it suitable for building scalable and high-performance applications. In FoodTrust, Node.js handles server-side logic, request processing, and communication between system components.

Database & Cloud Services

Firebase Firebase is a backend-as-a-service (BaaS) platform that provides real-time databases, authentication, cloud storage, and hosting services. In FoodTrust, Firebase is used for secure user authentication, real-time data synchronization, and storage of application-related data, reducing backend complexity and improving scalability.

Blockchain Technology

Blockchain (Ethereum / Polygon): Blockchain technology is used to ensure transparency, immutability, and trust within the FoodTrust system. Donation records, pickup confirmations, and delivery transactions are stored on the blockchain, preventing data tampering. Smart contracts automate reward distribution and verification processes, promoting accountability among stakeholders.

Solidity: Solidity is a smart contract programming language used to develop decentralized applications on Ethereum-compatible blockchains. In FoodTrust, Solidity is used to implement smart contracts for donation verification and token-based incentives.

Artificial Intelligence & IoT

Artificial Intelligence (AI) : AI models are used to predict food freshness and spoilage based on environmental parameters such as temperature and humidity. This helps NGOs prioritize food pickups and ensures food safety before distribution.

Internet of Things (IoT) : IoT sensors are used to monitor food storage conditions in real time. Temperature and humidity sensors collect environmental data, which is processed by the system to assess food quality and freshness.

Development & Testing Tools

Postman : Postman is an API development and testing tool used to design, test, and validate RESTful APIs. In FoodTrust, Postman is used to test backend services, ensure smooth data flow between components, and debug API responses efficient

6. METHODOLOGY

The FoodTrust project follows a systematic and structured methodology to design, develop, and evaluate a decentralized food donation and traceability system. The methodology ensures reliable implementation, efficient data handling, and seamless coordination between donors, NGOs, and administrators. The overall process is divided into multiple phases, as described below.

Project Methodology Phases

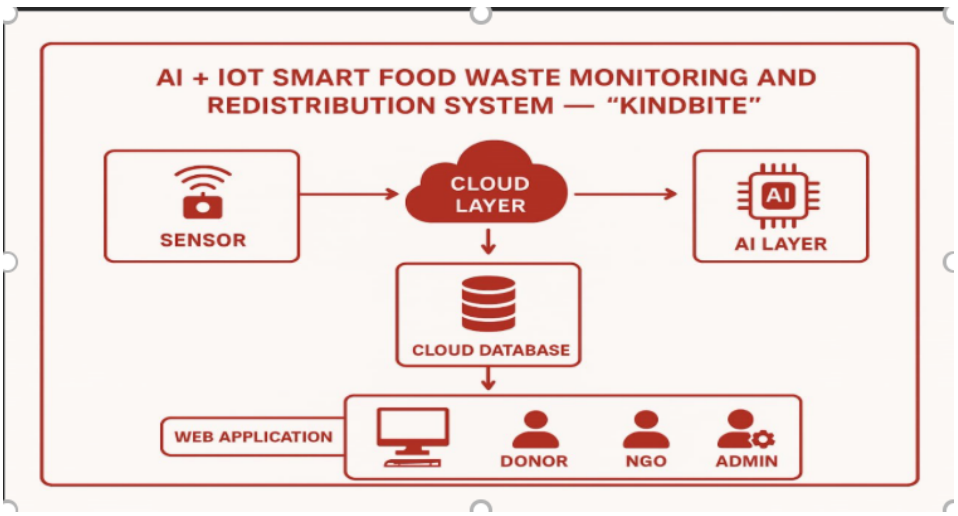
Phase	Module	Description	Technologies Used
Requirement Analysis	-	Identification of project objectives, stakeholder needs, system requirements, and scope of work.	Research & Analysis
Design	**UI/UX Module**	Design of wireframes, dashboards, and user workflows for donors, NGOs, and administrators.	Figma
Frontend Development	**Dashboard Module**	Development of interactive dashboards for donation listing, pickup tracking, and & results.	React.js, Tailwind CSS
Backend Development	**Authentication Module**	Implementation of secure user registration, login, and role-based access control.	Node.js, Express.js, Firebase Authentication
	Donation Management Module	Handling food donation submission, acceptance, and status updates.	Node.js, Express.js
	Sensor Data Module	Collection and processing of temperature and humidity data from IoT sensors.	IoT Sensors, MQTT/HTTP

Blockchain Integration	**Traceability Module**	Secure recording of donation, pickup, and delivery transactions using smart contracts.	Blockchain (Ethereum/Polygon), Solidity
AI Integration	**Freshness Prediction Module**	Analysis of sensor data to predict food freshness and spoilage risk.	Python, Machine Learning Models
Testing	**Testing Module**	Functional and API testing to ensure system reliability and accuracy.	Postman
Deployment	**Deployment Module**	Deployment of the application on a cloud-based hosting environment.	Cloud Hosting Services
Maintenance & Updates	**Maintenance Module**	Bug fixes, performance optimization, feature enhancement, and model improvements.	Ongoing Development

Methodological Workflow

1. **Data Collection:** Donation details and sensor data are collected through user inputs and IoT devices.
2. **Data Processing:** Sensor data is analyzed to assess food conditions and freshness levels.
3. **Verification:** Donation and delivery records are verified using blockchain smart contracts.
4. **Monitoring:** Administrators monitor system performance and generate reports.
5. **Optimization:** Continuous improvements are made based on feedback and system performance.

This methodology ensures a reliable, scalable, and transparent implementation of the FoodTrust system while maintaining food safety, trust, and operational efficiency



7. Expected Outcomes

The Expected Outcomes section describes the anticipated results, deliverables, and contributions of the FoodTrust project upon successful completion. The outcomes correspond directly to the project objectives and demonstrate the potential impact, usefulness, and applicability of the proposed solution in addressing food wastage and food insecurity. The FoodTrust system is expected to deliver a fully functional, deployable software product integrated with IoT, AI, and blockchain technologies that collectively ensure transparent, safe, and efficient redistribution of surplus food.

8.1 Primary Deliverable

The primary deliverable of the FoodTrust project is a deployable, multi-stakeholder web application that integrates blockchain-based traceability, IoT-based food condition monitoring, and AI-driven freshness prediction into a unified platform. The application will support three distinct user roles — Donor, NGO, and Administrator — each with dedicated dashboards and role-specific functionalities, accessible through an intuitive and responsive interface.

8.2 Functional Outcomes

The following functional outcomes are expected upon successful implementation of the FoodTrust system:

- **Donor & NGO Coordination Platform:** A fully operational web-based platform enabling donors to register, list surplus food with relevant details, and track donation status. NGOs will be able to browse available donations, accept pickup requests, and confirm final delivery, ensuring seamless end-to-end coordination.
- **Real-Time IoT Food Monitoring:** Integration of temperature and humidity IoT sensors that continuously monitor food storage conditions. Sensor data will be streamed to the platform dashboard in real time, allowing stakeholders to assess food quality before and during transportation.
- **AI-Based Freshness Prediction:** A trained machine learning model capable of predicting food freshness levels and spoilage risk using environmental sensor inputs. The model is expected to achieve a classification accuracy of 85% or higher, enabling data-driven pickup prioritization and reducing health risks.
- **Blockchain Transparency & Traceability:** Immutable, tamper-proof records of all donation transactions, pickup confirmations, and delivery completions stored on the Ethereum or Polygon blockchain. This ensures end-to-end traceability and builds trust among all stakeholders.
- **Automated Smart Contract Incentives:** Solidity-based smart contracts that automatically verify donation activities and distribute token-based rewards to active donors and NGOs, eliminating the need for manual verification and encouraging sustained platform participation.
- **Reduction in Food Wastage:** The platform is expected to contribute to a measurable decrease in surplus food wastage by enabling timely pickups and efficient redistribution. Simulated case studies will demonstrate the improvement in food redistribution rates compared to manual or uncoordinated systems.
- **User-Friendly Interface:** An intuitive, responsive, and accessible user interface designed for all stakeholders, including those with limited technical expertise. The interface is expected to achieve a System Usability Scale (SUS) score of 75 or above, indicating good usability.

8.3 Summary of Expected Outcomes

Table 1. Summary of Expected Outcomes for FoodTrust Project

Component / Objective	Expected Outcome	Measurable Deliverable
Donor & NGO Platform	A fully functional web-based platform enabling donors to list surplus food and NGOs to accept, schedule, and confirm pickups in real time.	Deployed multi-stakeholder web application with role-based dashboards for Donor, NGO, and Admin.
IoT Food Monitoring	Continuous real-time monitoring of food storage conditions (temperature & humidity) through integrated IoT sensors.	Working IoT sensor module transmitting live environment data to the platform dashboard.
AI Freshness Prediction	AI/ML model accurately predicting food freshness and spoilage risk based on sensor readings, enabling proactive pickup decisions.	Trained predictive model with ≥ 85% accuracy on freshness classification; integrated into backend API.
Blockchain Traceability	Immutable, tamper-proof records of all donation, pickup, and delivery transactions stored on-chain for end-to-end transparency.	Smart contracts deployed on Ethereum/Polygon testnet; full audit trail accessible via admin dashboard.
Smart Contract Incentives	Automated verification and token-based reward distribution for active donors and NGOs without manual intervention.	Solidity smart contracts with incentive logic; test results demonstrating correct automated reward execution.
Food Waste Reduction	Measurable reduction in surplus food wastage through timely redistribution facilitated by the platform.	Case study / simulation data showing reduction in food disposed vs. food redistributed.
User Experience	Intuitive, responsive, and accessible interface for all stakeholders including donors, NGOs, and admins.	Usability-tested UI/UX; positive SUS (System Usability Scale) score ≥ 75.

8.4 Technical and Social Impact

From a technical perspective, the FoodTrust project demonstrates the practical integration of three advanced technologies — blockchain, IoT, and AI — within a single cohesive system,

contributing to the fields of decentralized application development, smart contract engineering, and intelligent food safety analysis. The project validates the feasibility of applying distributed ledger technology to social welfare use cases beyond financial transactions.
From a social perspective, FoodTrust directly addresses the twin challenges of food wastage and food insecurity. By enabling safe, traceable, and accountable redistribution of surplus food, the platform supports underprivileged communities in gaining access to nutritious food while promoting responsible resource utilization among food service institutions. The system aligns with the United Nations Sustainable Development Goals (SDGs), particularly SDG 2 (Zero Hunger) and SDG 12 (Responsible Consumption and Production).

8.5 Future Scope

The FoodTrust platform has significant potential for future enhancement and scalability. Planned future developments include:

- Expansion to a mobile application (Android and iOS) to improve accessibility for donors and NGO field workers.
- Integration of advanced computer vision models for automated food quality assessment using camera-based inputs.
- Adoption of Layer 2 blockchain solutions to reduce transaction costs and improve throughput for large-scale deployments.
- Incorporation of route optimization algorithms to improve last-mile delivery efficiency for NGOs.
- Multilingual support to broaden the platform's reach across diverse communities and geographies.

In summary, the FoodTrust project is expected to deliver a technically robust, socially impactful, and commercially scalable solution for food donation and redistribution. The system demonstrates that emerging technologies can be effectively harnessed to address pressing humanitarian challenges, establishing FoodTrust as a meaningful contribution toward building a sustainable, transparent, and hunger-free society.

7. PROBLEM STATEMENT

Food wastage and food insecurity continue to be critical global challenges affecting millions of people. Large quantities of surplus edible food are discarded daily by restaurants, hotels, event venues, and institutional kitchens due to inefficient planning, lack of coordination, and absence of reliable redistribution mechanisms. At the same time, underprivileged communities face persistent food shortages, highlighting a severe imbalance between food availability and access.

Existing food donation and redistribution systems often rely on centralized platforms that lack transparency, real-time monitoring, and accountability. Donors are frequently uncertain about whether their donated food reaches beneficiaries safely and on time, while non-governmental organizations (NGOs) struggle with delayed pickups, food spoilage, and inefficient logistics. Furthermore, the absence of reliable food condition monitoring and traceability increases health risks and reduces trust among stakeholders.

To address these challenges, the **FoodTrust** project proposes a **decentralized food donation and traceability system** that integrates **blockchain, Internet of Things (IoT), and artificial intelligence (AI)** technologies. By enabling real-time monitoring of food conditions, predictive freshness analysis, and secure, tamper-proof transaction records, the system aims to ensure safe, transparent, and efficient redistribution of surplus food. FoodTrust seeks to minimize food wastage, enhance trust among donors and NGOs, and improve access to safe food for communities in need.

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